Example 6.8

The wholesale price of a commodity for seven consecutive days in a month is as follows:

Days	1	2	3	4	5	6	7
Commodity/price/ quintal	240	260	270	245	255	286	264

Calculate the variance and standard deviation.

Solution:

The computations for variance and standard deviation is cumbersome when x values are large. So, another method is used, which will reduce the calculation time. Here we take the deviations from an assumed mean or arbitrary value A such that $d = x - A$

In this question, if we take deviation from an assumed A.M. =255. The calculations then for standard deviation will be as shown in below Table;

Observations (x)	$d = x - A$	d^2
240	–15	225
260	5	25
270	15	225
245	–10	100
255 A	0	0
286	31	961
264	9	81
	35	**1617**

$$\text{Variance} = \sigma^2 = \frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2$$

$$= \frac{1617}{7} - \left(\frac{35}{7}\right)^2$$

$$= 231 - 5^2$$

$$= 231 - 25$$

$$= 206$$

$$\text{Standard deviation } \sigma = \sqrt{variance}$$

$$\sigma = \sqrt{206} = 14.35$$

Ch6_11th Stats.indd 167 2/3/2019 11:02:02 AM